

EDWARD KUBIAK

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Summary

Full stack developer and AI systems engineer. Creator of CAST v10.0.0 — a local-first, open-source multi-agent control plane for Claude Code whose 42-table execution record is searchable, signed, and predictive — shipped as a 9-package Homebrew ecosystem backed by 3,323 tests. Creator of Compute Atlas (compute-atlas.com) — an open, source-cited census of the U.S. grid-scale compute buildout spanning 1,231 facilities across 50 states — data centers, crypto mining, and dedicated power generation — published as open data with a public API. Builds deterministic, zero-LLM agent-reliability tools (misfire, attest, looptrip) shipped on PyPI and Homebrew. Ships and maintains production web applications at META Solutions across Ohio's K-12 education ecosystem. Open to full-stack and AI-infrastructure roles.

Skills

- **AI & Agent Systems:** Claude Code (hooks, agents, skills), Claude API, CAST v10.0.0 (27 agents, 9 packages), Agent Orchestration & Dispatch, Hook Architecture, MCP (Model Context Protocol), OpenTelemetry (GenAI spans), Deterministic Agent-Reliability Tooling, RAG / Embeddings (FTS5, Ollama), Managed Agents (beta)
- **Languages & Frameworks:** TypeScript, JavaScript (Node.js, ESM), Python, Rust, Bash, React 18/19, Next.js 16, Express 4/5, Vite, Tauri 2, Flask, Tailwind CSS v4, TanStack Query v5, Motion, MUI, AG Grid, xterm.js
- **Data:** SQLite (better-sqlite3, FTS5), PostgreSQL, MS SQL Server, MongoDB, BigQuery, SQL, Server-Sent Events (SSE)
- **Testing & Delivery:** Vitest, React Testing Library, BATS, pytest / unittest, GitHub Actions, Homebrew Packaging, shellcheck, Docker / Docker Compose, Traefik, Jenkins, launchd, Git

Open Source — AI Developer Tooling

Creator & Maintainer — CAST & Agent-Reliability Tools

2024 — Present

- Creator of CAST v10.0.0 "Make the Gates Tell the Truth" — a local-first, open-source multi-agent control plane for Claude Code: 27 specialist agents with hook-driven dispatch, model-aware routing, hook-enforced quality gates, and a 42-table SQLite execution record that acts: full-text search over every session (cast ask), signed SHA-256 audit receipts (cast ledger --verify), and telemetry-driven cost prediction (cast predict). v10 audited every quality gate against one question — what does this check report when the thing it guards did not happen? — and rebuilt the ones that reported success either way; new tests are mutation-tested against the bug they guard before they are trusted. 3,323 tests, zero cloud dependencies.
- Shipped the CAST ecosystem as 9 Homebrew packages plus the umbrella cast formula — the flagship framework, cast-desktop (native Tauri 2 + React 19 + Rust app with an embedded Express 5 + SQLite backend, 11 dashboard views, and a real PTY-backed terminal), the Claude Code Dashboard v2.7.0 (React 19 + TypeScript + Express 5 + SSE observability UI), and standalone packages for agent memory, health checks, journaling, MCP access, signed receipts, and dispatch prediction.
- Built a family of deterministic, zero-LLM agent-reliability tools, each shipped to PyPI and Homebrew with CI: misfire v0.2.0, a trace-grounded CLAUDE.md adherence auditor that finds which rules agents actually ignore; attest v0.3.0, a hook that verifies a subagent's DONE claim against the real git delta; and looptrip v0.1.2, a coordination-pathology detector that trips duplicate-work loops at iteration 2 and reproduces prevented duplicate-work spend from a committed fixture.

Creator — Compute Atlas

2026 — Present

- Built and operate Compute Atlas (v1.29.0) — an open, source-cited census of the U.S. grid-scale compute buildout covering data centers, crypto-mining sites, and the dedicated power generation contracted to feed them: 1,231 facilities across 50 states, 26.5 GW operational, 107.5 GW under construction and ~313.5 GW in the planned pipeline, every record carrying at least one public source and an explicit confidence level (confirmed / reported / rumored). Built solo on Next.js 16 · React 19 · TypeScript · MapLibre GL · Neon Postgres · Drizzle · Vercel.
- Shipped the full product: an interactive MapLibre GL map with water and geology overlays, sortable data tables, per-facility dossiers, and reference hubs (rankings, power, operators, metros, states, AI-vs-crypto), plus a /learn tier answering the underlying questions in cited prose and a public CORS-open JSON API — with the entire dataset released as open data (code MIT, data CC BY 4.0).
- Engineered an autonomous daily discovery pipeline — a single bounded `claude -p` run that discovers, enriches, and stages candidate facilities behind a fail-closed kill switch, a heartbeat, extraction guards, and a self-reverting cap, mechanically verifying every candidate source URL before staging, with human approval required before any record goes live.
- Extended the survey beyond capacity into its civic footprint — a community-opposition dimension recording litigation, moratoria, referendums and the projects opposition defeated, and a named-stakeholder dimension tying founders, owners, investors, executives, landowners and public officials to specific sites under a mandatory-citation, site-level evidentiary bar.

Professional Experience

Applications Developer · META Solutions — Columbus, OH

August 2022 — Present

- Spearheaded the complete migration of CrossCheck from AngularJS to React — a mission-critical EMIS data validation platform (React 18, AG Grid, MUI, TanStack Query, JWT auth) serving Ohio school districts.
- Architect and maintain production web applications across React, Flask, Express, PostgreSQL, and MS SQL Server, each serving Ohio's K-12 education ecosystem.
- Engineered SES-Wiki from the ground up — a React 19 + Express 5 EMIS scenario reference tool with JSON persistence, automated backups, and comprehensive Vitest coverage.
- Launched the Customization Web Store (CWS) — an internal React 19 + Vite platform that streamlined PowerSchool customization requests, replacing a manual email-based workflow.
- Modernized the E-Rate dashboard into a Docker Compose monorepo — dual React frontends, a Flask REST API with PostgreSQL, Traefik reverse proxy, and Jenkins CI/CD — helping districts capture federal telecom discount funding.
- Maintain and extend PowerSchool plugin customizations (jQuery/DataTables), including the Alert Builder notification system deployed across multiple client districts.
- Use Claude Code and CAST daily as production development infrastructure — the same multi-agent tooling published as open source.

Education

- Full Stack Web Development Certificate — The Ohio State University · 2022
- Bachelor of Arts and Science in Geological Science — Ohio University · 2005–2009